Lab 1 - Topic, Environment and Architecture

Childhood Vaccination, Socioeconomic Conditions, and Under-Five Mortality

Andrei Bujor | 4 November 2026

GitHub: AndreiBujor18/childhealth-explorer

1.2 / Topic and research question

How are childhood vaccination, socioeconomic conditions and access to basic services associated with under-five mortality across countries and over time? The project relates to Sustainable Development Goal 3 (SDG 3) by examining country-level differences in child health. It uses free annual data from UNICEF (vaccination) and the World Bank (mortality and socioeconomic indicators), initially covering 20 countries from 2000 to 2023. Students, public health analysts and NGOs will be able to compare countries, explore trends and estimate under-five mortality. The proposed machine learning (ML) task compares linear regression and random forest regression using the six explanatory indicators.

Planned data split: training 2000-2017, validation 2018-2020, testing 2021-2023. Preprocessing will be fitted on training data only. The models will estimate mortality for the same year as the input data.

1.3-1.4 / Development environment

Tool	Version / configuration
Python / Git	3.14.6 / 2.51.0.windows.1
VS Code / pgAdmin	1.143.0 / 9.18
PostgreSQL	18.6 portable; global_health; gh_user; UTF8
Python libraries	requests, pandas, SQLAlchemy, psycopg, FastAPI, Jinja2, pytest

The project uses the .venv virtual environment; dependency versions are pinned in requirements-lock.txt. VS Code has the Python, Jupyter and container extensions. PostgreSQL was extracted from the official EDB Windows ZIP into .local/pgsql, with data in .local/pgdata. It runs locally on 127.0.0.1:5432. Credentials are excluded from Git.

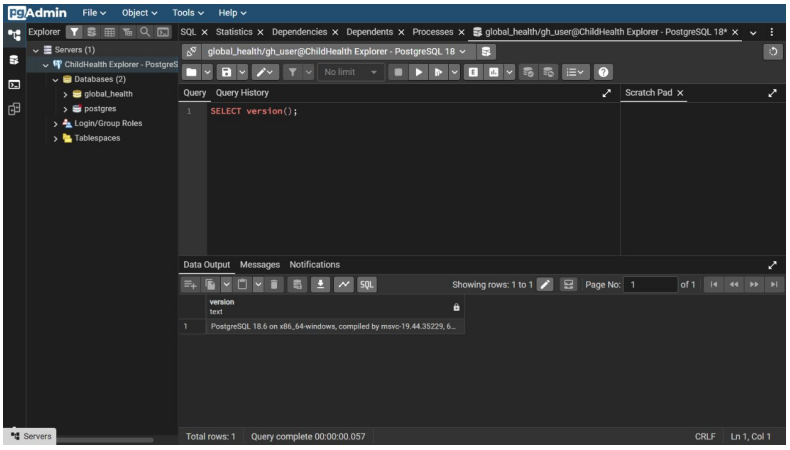


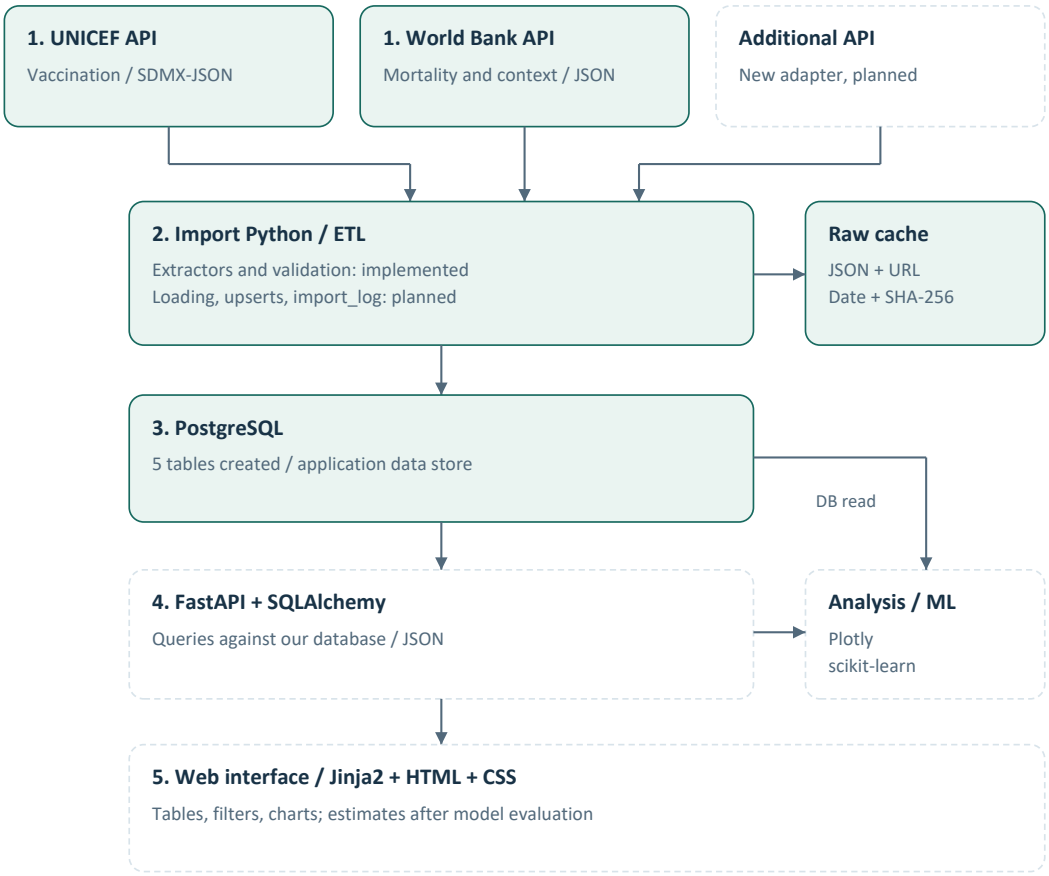
Figure 1. pgAdmin: global_health / gh_user, SELECT version(), PostgreSQL 18.6 (4 November). Full-size screenshots: docs/evidence/screenshots/.

1.5 / Topic-specific challenges

Challenge	Proposed solution
Different reference populations and ages	Keep DTP3, MCV1 and under-five mortality separate. Preserve definitions, units and age groups; align records by country and year.
Confounding in the vaccination-mortality relationship	Include GDP, health spending, water and sanitation. Discuss unmeasured factors and interpret results as associations.

1.6 / Application architecture

ChildHealth Explorer / application architecture



Only the extract, transform and load (ETL) pipeline will retrieve external data and load observations. The backend and analysis modules will read PostgreSQL; the interface will use the backend. This keeps browsing independent of external API availability. Source adapters, validation and the empty database are implemented. Data loading and the components in dashed boxes are planned.

Integrating a third API requires a new adapter, registry entry and indicator configuration. The adapter will return the same fields: source, indicator code, ISO3 country code, year, sex and value. The existing validation can be reused. Additional data dimensions would require a schema review.

Folders: app/ - database and models; etl/ - adapters, cache and validation; config/ - indicator catalog; db/ - schema and export; docs/ - design and evidence; scripts/ - setup; notebooks/ - analysis; tests/ - checks. Planned UI: Jinja2 and Plotly; ML: scikit-learn.

1.7 / Indicators and data availability

Source / code	Indicator and unit	Non-null values
UNICEF / IM_DTP3	DTP3 vaccination coverage (%)	480 / 480
UNICEF / IM_MCV1	First-dose measles vaccination coverage (%)	480 / 480
WB / SH.DYN.MORT	Under-five mortality per 1,000 live births	480 / 480
WB / NY.GDP.PCAP.PP.KD	GDP per capita, PPP; constant 2021 international $	480 / 480
WB / SH.XPD.CHEX.GD.ZS	Current health expenditure (% of GDP)	480 / 480
WB / SH.H2O.BASW.ZS	At least basic drinking water services (% of population)	470 / 480
WB / SH.STA.BASS.ZS	At least basic sanitation services (% of population)	479 / 480

Under-five mortality is the target; the other six indicators are proposed explanatory variables. The pilot covers 20 countries and 24 years, using data for both sexes combined where applicable. **469 of 480 country-year combinations contain all seven values.** These data establish feasibility; the countries were not selected to form a statistically representative sample.

Missing values remain NULL. Adapters normalize ISO3 codes and response structures; validation checks years, duplicates and numeric limits. Pagination, retries and a raw JSON cache support reliable extraction. Each cached response has a source URL, retrieval timestamp and SHA-256 hash.

API verification

All seven indicator URLs for Romania returned HTTP 200 and 24 annual records each on 3 November. On 4 November, all seven were opened in Chrome and their JSON responses were confirmed. World Bank returns pagination metadata followed by observations; UNICEF uses SDMX dimensions and indexed observations. The browser screenshots and automated response checks are stored separately.

The complete clickable URL catalog is in docs/indicators.md. The 21 October pilot is recorded in docs/evidence/api_validation.json, coverage.csv and sample_observations.csv. The later Romania checks are in url-check-2026-11-03.json and browser-url-check-2026-11-04.json in the same folder. These checks save files without inserting data into PostgreSQL.

Source documentation

UNICEF / SDMX API and response structure

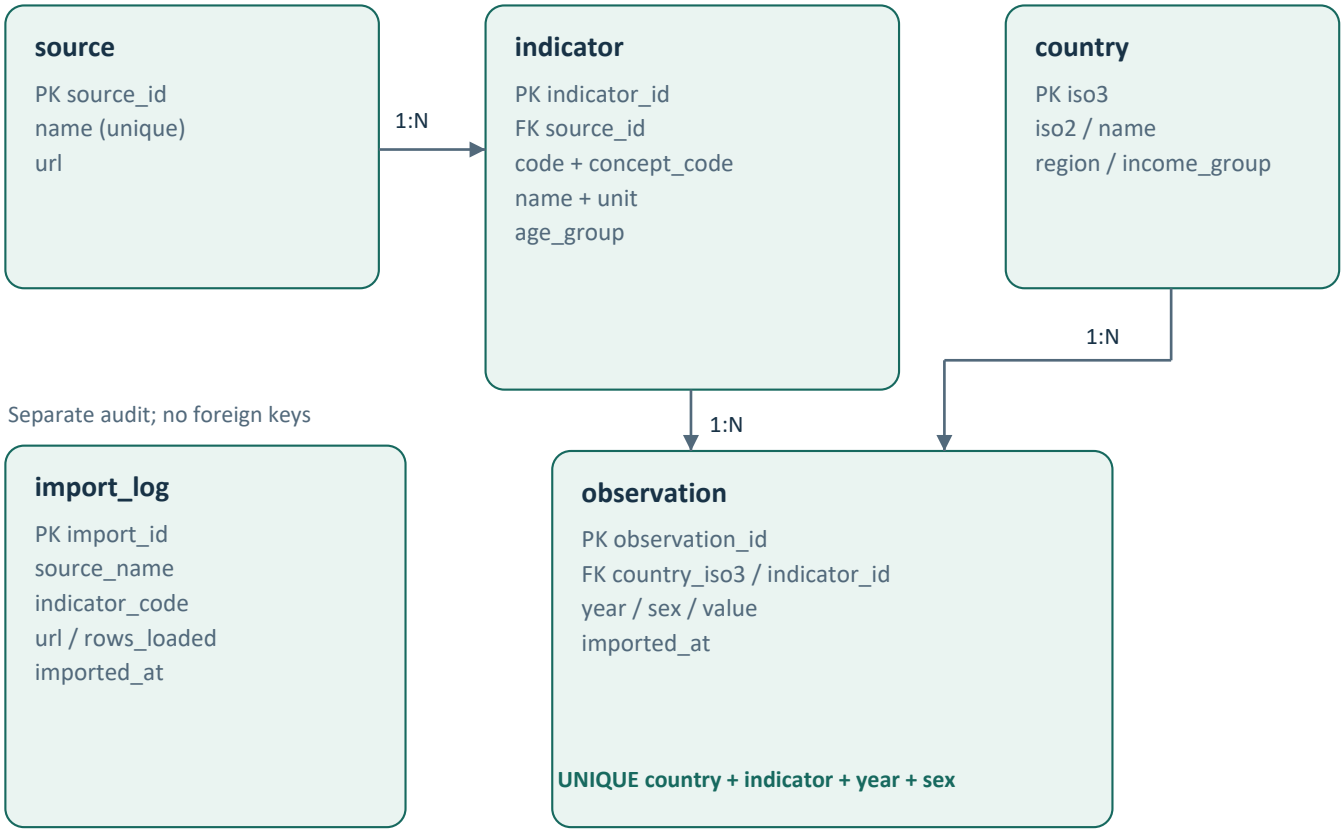
UNICEF / Immunization indicators and definitions

World Bank / Indicators API

EDB / PostgreSQL ZIP binaries

1.8-1.9 / Database design and implementation

Database / entity-relationship model



Justified changes to the reference schema

Source-specific codes: UNIQUE(source_id, code) allows different APIs to use the same code. **concept_code:** identifies a shared concept without assuming equivalent definitions. **age_group:** records the fixed reference population of each indicator. Code and unit fields have been expanded to accommodate longer metadata.

Observations remain unique by country, indicator, year and sex; the indicator identifies the source. Missing values remain NULL. The (indicator_id, year) index is retained. Check constraints enforce years 1900-2100, valid sex codes and nonnegative values. Percentage limits are also checked in Python.

Implementation and verification

The five tables are defined by SQLAlchemy models in app/models.py and created by app/create_tables.py. The schema was exported to db/schema_dump.sql using pg_dump --schema-only. An automated comparison against db/schema.sql confirmed matching columns, keys, constraints and indexes.

A read-only check on 4 November confirmed that all five tables exist and are empty, as required for Lab 1. Evidence: docs/evidence/database_verification.json and recheck-2026-11-04.json.

import_log is a separate audit table without foreign keys, following the lab design. Database loading, application endpoints, a database-connected interface and model training will be implemented in later stages.